

Pranay Kashyap

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EDUCATION

- **New York University** Brooklyn, NY
Master of Science in Applied Data Science Sep. 2024 – Present
- **National Institute of Technology** Srinagar, India
Bachelor of Technology in Information Technology Aug. 2018 – July. 2022

SKILLS

- **Languages & Frameworks:** Python, Java, JavaScript, Kotlin, C/C++, SQL, PyTorch, TensorFlow, ONNX, Keras, Detectron2, MMDetection, YOLO, Scikit-learn, Pandas, SpaCy, NLTK, Gensim, NumPy, OpenCV
- **Tools & Platforms:** FastAPI, Flask, Docker, Triton Server, WebSockets, CI/CD, AWS, FFmpeg, Android SDK/NDK, Jetpack, Firebase, Gradle, Git, CUDA, Anaconda, Linux

EXPERIENCE

- **Kaliber AI** San Mateo, CA
Machine Learning Intern May 2025 - Dec 2025
 - Developed a real-time medical voice assistant with a modular FastAPI backend, WebSocket streaming, speaker-verified Voice Activity Detection, and optimized Whisper-large-v3, enabling partial-text transcription and achieving sub-2s latency.
 - Engineered interrupt-driven TTS and dynamic buffering to reduce perceived latency by 35% and improve conversational flow.
 - Containerized and deployed full AI stack on AWS EC2 (g5.xlarge) and Triton Inference Server, leveraging GPU acceleration, TensorRT-optimized models, and Dockerized microservices for reproducible, zero-downtime production deployments.
 - Built scalable microservice CI/CD architecture with automated GitHub workflows, dynamic model loading, and browser-accessible interface, enabling rapid, independent module updates and 3× faster LLM inference.
- **Samsung Research** Bengaluru, KA
Senior Software Engineer Jul 2022 - Aug 2024
 - Led the implementation of a DNN model for Generic Event Boundary Detection (GEBD), now a core component of the Video Moments solution in the Enhance-X app. This app, available on the Galaxy Store, processed data on-device in real-time, helping users efficiently identify and manage key moments in their videos.
 - Boosted accuracy in video event captioning by 20% by modifying CLIP to incorporate optical flow and pixel movement, capturing temporal dynamics for precise boundary-aware captions.
 - Developed a zero-cost approach for GEBD using the Segment Anything Model (SAM), achieving state-of-the-art results, and publishing the work at ICASSP'24, this method repurposed SAM for class-agnostic video boundary detection.
 - Optimized GEBD model through ONNX-based quantization, cutting size by 71.4% and accelerating edge deployment with sustained KPI performance and reduced computational overhead.
 - Enhanced backend performance by integrating scalable APIs using FFmpeg for real-time video frame processing.
 - Built a proof-of-concept (PoC) Android app showcasing the Video Screenshot use case, enabling real-time frame extraction and on-device model inference.
- **Samsung Research** Bengaluru, KA
Software Engineer Intern Jan 2022 - Jul 2022
 - Streamlined GEBD by retraining BSN and BMN models, cutting computational complexity by 80%.
 - Developed a self-similarity-based 3D filter pruning method to compress CNNs without post-training finetuning, achieving 12% faster inference and 10% model size reduction on video models, enabling deployment on edge devices.
 - Applied 3D filter pruning to X3D in the GEBD pipeline, cutting size and speeding inference, with near-zero accuracy drop.
 - Built CI-ready unit and integration tests to ensure model reliability and cross-platform deployment.
- **Tolstoy** San Francisco, CA
Machine Learning Intern May 2020 - Aug 2020
 - Designed a multi-class text classification model (Bitesize) to deliver accurate categorization of text messages.
 - Designed an end-to-end OCR and Name Entity Recognition (NER) pipeline to streamline Harvard specimen label digitization.
 - Deployed models as interactive web applications using Streamlit, simplifying user interaction.
 - Created an API for the Bitesize model using Flask, ensuring scalability and efficient deployment.

RESEARCH PAPERS & PATENTS

- SAMnet: Semantic Aware Multimodal Network for Emoji Drawing Classification — CVIP'22
- Repetition Counting from Compressed Videos Using Sparse Residual Similarity — ICASSP'23
- Self-Similarity is All You Need for Fast and Lightweight GEBD — ICASSP'23
- Whats in the Flow Exploiting Temporal Motion Cues for Unsupervised GEBD — WACV'24
- SAM-GEBD: Zero-Cost Approach for Generic Event Boundary Detection — ICASSP'24
- Method and electronic device for generating a segment of a video (2023)
- Generating contextual transition effects in videos (2024)
- Electronic device and method for adaptive video display management based on region localized regeneration of frames (2024)

AWARDS

- **LM-Studio On-Device AI Builders Hackathon'24** : Built an on-device app using Donut and LLaMA models.
- **Samsung Best Paper Award 2024 - Bronze** : Awarded for paper on physics-based interactive dynamics.
- **Samsung IFest'23 Ideation Contest** : For creating breakthrough idea and showcasing PoC on VideoAI.
- **Hackerearth Machine Learning Challenge - Predict severity of plane crash** : World rank 10